

- 37** Emphysema is a type of chronic obstructive pulmonary disease, COPD. People with emphysema have a very low level of a plasma protein which inhibits the enzyme elastase.

Elastase breaks down the elastic fibres in the bronchioles and alveoli.

Which are effects of the low levels of inhibitor in people with emphysema?

- 1 alveoli do not stretch and recoil properly during inhalation and exhalation
- 2 blood is poorly oxygenated resulting in a rapid breathing rate
- 3 bronchioles collapse during exhalation trapping air in the alveoli

A 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only